

DIREC: Diffusion-Based Review-Embedding Generation for Accurate Cross-Domain Recommendation

Jiwon Son
tinybeing@hanyang.ac.kr
Hanyang University
Seoul, Korea

Yongjin Kwon
maruci1@hanyang.ac.kr
Hanyang University
Seoul, Korea

Sang-Wook Kim*
wook@hanyang.ac.kr
Hanyang University
Seoul, Korea

Abstract

The goal of Cross-domain Recommender System (CDRS) is to recommend items in a target domain for users who have no target-domain interactions by leveraging their source-domain interaction histories. Most existing CDRSs transfer a user embedding from the source domain to the target domain and predict ratings via embedding matching with target-domain item embeddings, which can overlook fine-grained user–item preference signals expressed in reviews. To capture such fine-grained signals for each target-domain user–item pair, we propose DIREC, a conditional-diffusion-based CDRS that generates a target-domain review embedding and then predicts the corresponding rating from the generated embedding. DIREC improves review-embedding generation with two key ideas: (Idea 1) target-aware source attention to construct review guidance (*i.e.*, a conditioning embedding); and (Idea 2) pretraining on target-domain review embeddings from target-only users to learn a broader target-domain review-embedding distribution. Extensive experiments on three cross-domain scenarios show that DIREC substantially outperforms nine competitors, reducing MAE by up to 14.2%.

CCS Concepts

• **Information systems** → **Data mining**.

Keywords

Cross-domain Recommendation, Cold-start Recommendation, Review Embedding Generation, Diffusion Models

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Resource Availability:

The source code of this paper is available at https://github.com/Bigdasgit/DIREC and archived at https://doi.org/xxxx.

*Corresponding author.



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1 Introduction

Recommender systems enhance user satisfaction on online platforms by recommending items that match users' preferences. As online platforms continue to grow, users increasingly consume items across multiple domains (*e.g.*, books, music, clothing, and toys) [7, 10, 27]. Consequently, recommending to a user the items in her *new domain* has become increasingly important, making *Cross-Domain Recommendation (CDR)* attract growing attention [17, 31, 39, 45]. CDRSs aim to provide accurate recommendations in a *target domain* where a user has no observed interactions by leveraging her rich interaction history from a *source domain*.

Most existing CDRSs learn to transfer user embeddings from the source domain to the target domain, typically by utilizing *overlapping users* observed in both domains [14, 18, 26, 34, 44, 45]. They are generally trained using two objectives: (i) embedding alignment, which encourages the transferred source-domain embedding to be close to the corresponding target-domain embedding for each *overlapping user*, and (ii) rating prediction via user–item embedding matching in the target domain. Consequently, CDRSs can recommend items in the target domain for test users whose preference is unknown in the target domain, using only their source-domain interaction histories. Early work typically used MLP-based mapping functions for embedding alignment [26], while recent studies have explored meta-learning-based approaches [45] and conditional-diffusion-based approaches [16, 18, 37].

Meanwhile, *reviews* contain rich information about users and items, such as users' preferences and item characteristics, providing an important signal for recommendation. Some CDRSs additionally exploit textual reviews to build user and item embeddings [8, 19]. Specifically, they initialize user embeddings (resp. item embeddings) for each domain by aggregating the embeddings of reviews written by the user (resp. reviews written for the item) in that domain. As in conventional CDRSs, they then learn to transfer source-domain user embeddings into target-domain user embeddings using a mapping function. By leveraging the reviews to initialize user and item embeddings, these review-based CDRSs often achieve better performance than non-review-based ones [8, 43].

Beyond user- or item-specific information, reviews also include *fine-grained user–item preference signals* that are not fully captured by scalar ratings; for example, even when two users assign the same rating to an item, the aspects they like or dislike may differ. When trained only to predict ratings directly by matching user and item embeddings in the target domain, as in most existing CDRSs, a model may fail to sufficiently exploit the fine-grained user–item preference signals expressed in reviews for accurate recommendation.

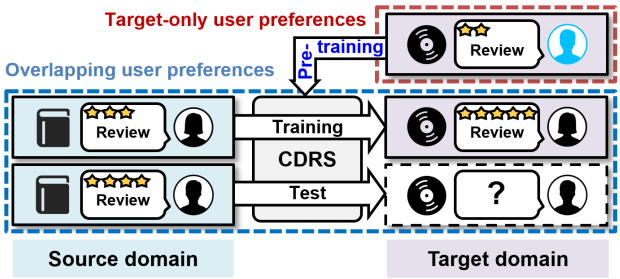


Figure 1: Illustration of the CDR setting. Existing CDRSs learn to transfer overlapping users' source-domain preferences (blue) to their target-domain preferences (light purple).

To capture these fine-grained signals more precisely for each pair of a user and an item in the target domain, we propose a two-stage approach that (i) generates a target-domain review embedding for the pair and then (ii) predicts the corresponding rating from the generated embedding. Specifically, we first obtain an item embedding in the target domain by aggregating the review embeddings written for the item. We then train the model to generate a review embedding for a pair of a user and an item in the target domain by utilizing her source-domain review embeddings and the target-domain item embedding. For the accurate generation of the review embedding in the target domain, we adopt a conditional diffusion model, which has been widely used in generative applications [1, 6, 23, 32, 37]. Conditional diffusion models are trained to denoise noisy samples, thereby modeling the underlying data distribution [3, 42]. Given a *condition embedding*, they can generate such samples well-aligned with the condition [13, 42]. There are two key challenges to generate high-quality samples: (i) constructing a condition embedding that effectively represents the desired samples and (ii) accurately modeling the target data distribution [40]. To address these two challenges, we propose two ideas that further improve the conditional diffusion model in our review-based CDRS: **(Idea 1)** target-aware source attention that constructs the *review guidance* (*i.e.*, a condition embedding for diffusion) by emphasizing source-domain reviews more relevant to the target-domain item; and **(Idea 2)** pre-training on target-domain review embeddings from target-only users to make the model understand target-domain review-embedding distribution more accurately.

(Idea 1) To generate a review embedding, we construct review guidance as a condition embedding by leveraging the user's source-domain review embeddings and the target-domain item embedding. Conditioned on this review guidance, the diffusion model generates the embedding of a review corresponding to a specific user–item pair. However, some of the user's source-domain reviews may be weakly related to a given target domain item. They are unlikely to contribute to generating the target-domain review embedding; therefore, we should carefully aggregate the user's source-domain review embeddings when constructing review guidance. To this end, we propose target-aware source attention, which computes a weighted average of the user's review embeddings in the source domain, where the weights given with the similarities between the target domain item and the source-domain items that she interacted with. As a result, the proposed attention assigns higher weights to the review embeddings that are more relevant to the target domain item, yielding more informative review guidance.

(Idea 2) Generative models, such as diffusion models, generate data by sampling from the data distribution learned during training, either unconditionally [12] or conditionally given a specified condition [13]. Therefore, generating high-quality samples requires sufficient training data to model the target data distribution broadly [24]. As illustrated in Figure 1, existing CDRSs [2, 8, 18, 20] rely on training data *only from the overlapping users* who have preference information (*e.g.*, review) in both source and target domains. However, the target domain often contains much more information from *target-only users*, who do not have any interactions in the source domain, than from overlapping users. For example, in the Amazon Book (source domain) to Music (target domain) scenario, target-domain reviews written by overlapping users account for only about 34.5% of all reviews; statistics for other scenarios are reported in Table 2 in Section 4. To generate high-quality review embeddings, the diffusion model should learn the review-embedding distribution in the target domain from sufficient data; toward this end, we perform the pre-training of our diffusion model by leveraging the reviews of target-only users. This pre-training allows the diffusion model to better capture the review-embedding distribution and generate higher-quality review embeddings in the target domain.

We propose **DIREC**, an improved **DI**ffusion-based **RE**view-embedding generation model for **C**ross-domain recommendation that incorporates the two key ideas. DIREC consists of three main components: (C1) a review guidance encoder, (C2) a review embedding generator, and (C3) a rating predictor. (C1) constructs review guidance to condition the diffusion model. To this end, it employs (Idea 1) target-aware source attention, which aggregates the user's source-domain review embeddings by considering the relevance between the corresponding source-domain items and the target-domain item. (C2) generates the review embedding for the given user–item pair using a conditional diffusion model. The diffusion model is conditioned on the review guidance produced by (C1). (C3) is an MLP-based predictor that takes the generated review embedding as input and predicts the rating.

To generate review embeddings effectively and predict ratings accurately, DIREC is trained in two stages: (T1) pre-training on target-only users and (T2) fine-tuning on overlapping users. In (T1), DIREC is trained to unconditionally generate target-domain review embeddings for all target-only users from noise embeddings, enabling the diffusion model to better capture the target-domain review-embedding distribution (Idea 2). In (T2), DIREC is trained to generate the review embedding for the given user–item pair from a noise embedding where the model is trained with and without the review guidance by randomly dropping the guidance (classifier-free guidance [13]). In both (T1) and (T2), DIREC is trained end-to-end to predict the rating that the user would assign to the item based on the generated review embedding.

Our contributions are summarized as follows:

- **Important Claim.** We claim that reviews capture more fine-grained user–item preference information than ratings, and that learning to generate review embeddings is crucial for accurate rating prediction in cross-domain recommendation.

Table 1: Key notations used in this paper

Notation	Description
$\mathcal{U}^s, \mathcal{V}^s$	Sets of all users/items in the source domain
$\mathcal{U}^t, \mathcal{V}^t$	Sets of all users/items in the target domain
$\mathcal{U}^o$	A set of overlapping users ($\mathcal{U}^s \cap \mathcal{U}^t$)
$u_i^{s\setminus o}, u_i^{t\setminus o}$	A source-only user and a target-only user
u_i^s, v_j^s	The i-th user in $\mathcal{U}^s$ and the j-th item in $\mathcal{V}^s$
u_i^t, v_j^t	The i-th user in $\mathcal{U}^t$ and the j-th item in $\mathcal{V}^t$
u_i^o	The i-th user in $\mathcal{U}^o$
$\mathcal{X}^s, \mathcal{X}^t$	Sets of review embeddings in each domain
$\mathbf{x}_{ij}^s, \mathbf{x}_{ij}^t$	A review embedding written by user u_i for item v_j in each domain
$\mathcal{R}^s, \mathcal{R}^t$	Sets of observed ratings in each domain
r_{ij}^s, r_{ij}^t	A rating of user u_i for item v_j in each domain
$\hat{r}_{ij}^t$	A predicted target-domain rating for $(u_i^{s\setminus o}, v_j^t)$
$\mathbf{h}_{ij}$	A review guidance for (u_i, v_j)
$\hat{\mathbf{x}}_{ij}^t$	A generated target-domain review embedding for (u_i, v_j^t)

- **Novel Framework.** We propose DIREC, a novel conditional-diffusion-based CDRS that generates review embeddings and then predicts ratings based on the generated embeddings. We further introduce two ideas to more improve the recommendation accuracy of DIREC:
  - **(Idea 1)** DIREC constructs review guidance via target-aware source-review attention to condition the diffusion model for review-embedding generation.
  - **(Idea 2)** DIREC pre-trains the diffusion model on review embeddings from target-only users, enabling it to better cover the target-domain review-embedding distribution.
- **Extensive Experiments.** To validate the effectiveness of DIREC, we conduct extensive experiments using three cross-domain scenarios (*i.e.*, datasets) and nine competitors. Our results show that DIREC outperforms the best competitor significantly and consistently, reducing MAE by up to 14.2%.

2 Notations and Problem Definition

In this section, we first define the notations used in this paper and then formulate the cross-domain recommendation (CDR) problem.

Notations. Let $\mathcal{U}^s$ and $\mathcal{U}^t$ denote the sets of users in the source and target domains, respectively. $\mathcal{U}^o = \mathcal{U}^s \cap \mathcal{U}^t$ denotes the set of overlapping users u_i^o that appear in both domains. A *source-only user* $u_i^{s\setminus o} \in \mathcal{U}^s \setminus \mathcal{U}^o$ is observed only in the source domain (*i.e.*, unobserved in the target domain), while a *target-only user* $u_i^{t\setminus o} \in \mathcal{U}^t \setminus \mathcal{U}^o$ is observed only in the target domain. $\mathcal{V}^s$ denotes the set of items v_j^s in the source domain, and $\mathcal{V}^t$ denotes the set of items v_j^t in the target domain. $\mathcal{X}^s$ represents the set of review embeddings $\mathbf{x}_{ij}^s$ for all observed user–item pairs (u_i^s, v_j^s) in the source domain, and $\mathcal{X}^t$ represents the set of review embeddings $\mathbf{x}_{ij}^t$ for all observed user–item pairs (u_i^t, v_j^t) in the target domain; all review texts are encoded by using a pre-trained Sentence-Transformer.[1] $\mathcal{R}^s$ represents the set of observed ratings r_{ij}^s that user u_i^s assigned to source-domain items v_j^s, and $\mathcal{R}^t$ denotes the set of observed ratings r_{ij}^t that user u_i^t assigned to target-domain items v_j^t. Each r_{ij}^t takes a value in $\{1, 2, 3, 4, 5\}$. For a source-only user $u_i^{s\setminus o}$, both $\mathbf{x}_{ij}^t$ and r_{ij}^t are unknown (NULL).

Problem Definition. The goal of CDR is to predict the rating $\hat{r}_{ij}^t$ that a source-only user $u_i^{s\setminus o}$ would give to a target-domain item v_j^t. Based on the above notations, we define the problem as follows:

- **Given:** the source-domain review-embedding set $\mathcal{X}^s$, the target-domain review-embedding set $\mathcal{X}^t$, the observed source-domain rating matrix $\mathcal{R}^s$, the observed target-domain rating matrix $\mathcal{R}^t$, and (a source-only user $u_i^{s\setminus o}$, a target-domain item v_j^t).
- **Output:** the predicted rating $\hat{r}_{ij}^t$.

Table 1 summarizes the key notations used in this paper.

3 The Proposed Method

In this section, we elaborate on DIREC, a novel cross-domain recommendation model that exploits target-domain review embeddings generated by conditional diffusion for accurate rating prediction.

3.1 Overview

Figure 2 presents the overview of DIREC. Given a user-item pair $(u_i^{s\setminus o}, v_j^t)$, DIREC predicts the rating $\hat{r}_{ij}^t$ with three main components: (C1) a review guidance encoder, (C2) a review embedding generator, and (C3) a rating predictor.

First, (C1) constructs a review guidance $\mathbf{h}_{ij}$ by aggregating the source-domain review embeddings, $\{\mathbf{x}_{ik}^s\}_{k\in\mathcal{V}_i^s}$ of $u_i^{s\setminus o}$, by considering the similarity between each source-domain item v_k^s and the target-domain item v_j^t as the aggregation weight, where $\mathcal{V}_i^s$ denotes a set of items which $u_i^{s\setminus o}$ interacted with in the source domain. (C2) starts from a noise embedding $\mathbf{x}_{ij}^{t,(N)}$ and applies the reverse process for N steps, conditioned on the review guidance $\mathbf{h}_{ij}$, to generate the review embedding $\hat{\mathbf{x}}_{ij}^t$ (*i.e.*, $\hat{\mathbf{x}}_{ij}^{t,(0)}$). (C3) predicts the rating $\hat{r}_{ij}^t$ by feeding the generated review embedding $\hat{\mathbf{x}}_{ij}^t$ into an MLP-based predictor. DIREC is trained in two stages: (T1) pre-training on the target-domain review embeddings from *target-only users* to better capture the target-domain review-embedding distribution, and (T2) fine-tuning on the target-domain review embeddings from *overlapping users* to accurately generate the review embedding $\hat{\mathbf{x}}_{ij}^t$.

3.2 Inference Procedure

In this section, we describe the inference phase of DIREC, which consists of three main components.

3.2.1 Review guidance encoder. To construct the review guidance $\mathbf{h}_{ij}$, we first obtain the item embedding $\mathbf{v}_k^s \in \mathbb{R}^d$ for a source-domain item v_k^s by averaging its review embeddings $\{\mathbf{x}_{lk}^s\}_{l\in\mathcal{U}_k} \in \mathbb{R}^{|\mathcal{U}_k|\times d}$. Here, $\mathcal{U}_k$ denotes the set of users u_l who interacted with item v_k^s, and d denotes the dimensionality of the review embedding in the pre-trained sentence transformer. Similarly, we obtain the item embedding $\mathbf{v}_j^t \in \mathbb{R}^d$ for the target-domain item v_j^t by averaging its review embeddings $\{\mathbf{x}_{lj}^t\}_{l\in\mathcal{U}_j} \in \mathbb{R}^{|\mathcal{U}_j|\times d}$. Next, to obtain the review guidance $\mathbf{h}_{ij} \in \mathbb{R}^d$, we design a target-aware source attention mechanism. The target-aware source attention is implemented via

[1] https://huggingface.co/sentence-transformers/all-mpnet-base-v2

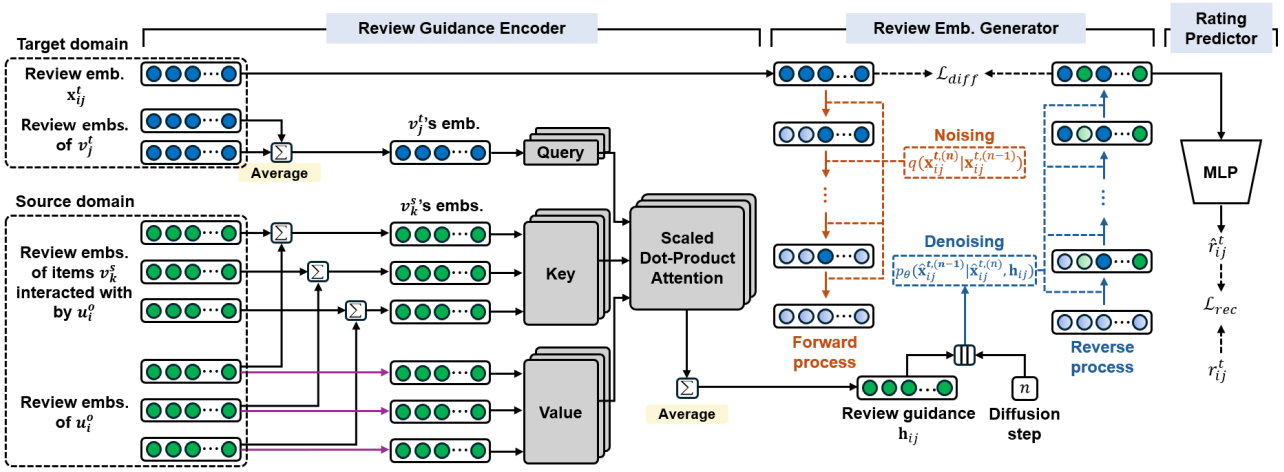


Figure 2: Overview of DIREC consisting of (C1) Review Guidance Encoder, (C2) Review Embedding Generator, and (C3) Rating Predictor. DIREC is trained in two stages: (T1) pre-training (C2) and (C3) on target-only users, and (T2) fine-tuning (C2) and (C3) on overlapping users by conditioning (C2) on the guidance from (C1) to generate $\hat{\mathbf{x}}_{ij}^t$ and predict $\hat{r}_{ij}^t$.

multi-head attention [33], where the target-domain item embedding $\mathbf{v}_j^t$ is used as a *query*, the embeddings of the items that the target user $u_i^{s\setminus o}$ interacted with in the source domain, $\{\mathbf{v}_k^s\}_{k\in\mathcal{V}_i^s}$, are used as *keys*, and the embeddings of the corresponding source-domain reviews written by $u_i^{s\setminus o}$ are used as *values*:

$$\mathbf{h}_{ij} = \mathrm{MHA}\left(\mathbf{v}_j^t, \{\mathbf{v}_k^s\}_{k\in\tilde{\mathcal{V}}_i^s}, \{\mathbf{x}_{ik}^s\}_{k\in\tilde{\mathcal{V}}_i^s}\right), \tag{1}$$

where $\mathrm{MHA}(Q,K,V)$ denotes multi-head attention, and Q, K, and V represent the query, key, and value, respectively. This constructs $\mathbf{h}_{ij}$ as a weighted aggregation of $\mathbf{x}_{ik}^s$, where the weights reflect the relevance between the target-domain item embedding $\mathbf{v}_j^t$ and each source-domain item embedding $\mathbf{v}_k^s$. To improve efficiency, we sample H items from $\mathcal{V}_i^s$; $\tilde{\mathcal{V}}_i^s \subseteq \mathcal{V}_i^s$ denotes this sampled set with $|\tilde{\mathcal{V}}_i^s| = H$. We report a sensitivity analysis on H in Section 4.

3.2.2 Review embedding generator. To generate the review embedding $\hat{\mathbf{x}}_{ij}^t$, (C2) performs the reverse process of the diffusion model conditioned on $\mathbf{h}_{ij}$. Since the forward process is used only during training, we describe it in Section 3.3 and focus only on the reverse process here.

During inference, the diffusion model's reverse process aims to generate the review embedding $\hat{\mathbf{x}}_{ij}^t \in \mathbb{R}^d$ by denoising an initial noisy embedding $\hat{\mathbf{x}}_{ij}^{t,(N)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ for N steps, producing $\hat{\mathbf{x}}_{ij}^{t,(0)}$. Here, $\mathcal{N}(\boldsymbol{\mu}_\theta(\cdot), \Sigma)$ denotes a Gaussian distribution with mean $\boldsymbol{\mu}_\theta(\cdot)$ and covariance Σ.

At step $n \in \{1,\dots,N\}$, conditioned on the review guidance $\mathbf{h}_{ij}$ obtained from (C1), we perform the reverse process in the same manner as the Denoising Diffusion Implicit Model (DDIM) [30]:

$$p_\theta\left(\hat{\mathbf{x}}_{ij}^{t,(n-1)} \mid \hat{\mathbf{x}}_{ij}^{t,(n)}, \mathbf{h}_{ij}\right) = \mathcal{N}\left(\hat{\mathbf{x}}_{ij}^{t,(n-1)};\ \boldsymbol{\mu}_\theta\left(\hat{\mathbf{x}}_{ij}^{t,(n)}, \mathbf{h}_{ij}, n\right),\ \Sigma\right). \tag{2}$$

To ensure stable generation of the review embedding and reliable rating prediction, we set $\Sigma = \mathbf{0}$, making the reverse process deterministic. Accordingly, we obtain the denoised review embedding $\hat{\mathbf{x}}_{ij}^{t,(n-1)}$ as follows:

$$\hat{\mathbf{x}}_{ij}^{t,(n-1)} = \boldsymbol{\mu}_\theta\left(\hat{\mathbf{x}}_{ij}^{t,(n)}, \mathbf{h}_{ij}, n\right), \tag{3}$$

$$\boldsymbol{\mu}_\theta\left(\hat{\mathbf{x}}_{ij}^{t,(n)}, \mathbf{h}_{ij}, n\right) = \left(\sqrt{\bar{\alpha}_{n-1}} - \sqrt{1-\bar{\alpha}_{n-1}}\frac{\sqrt{\bar{\alpha}_n}}{\sqrt{1-\bar{\alpha}_n}}\right) \times \mathrm{MLP}_{\mathrm{dif}}\left(\hat{\mathbf{x}}_{ij}^{t,(n)}, \mathbf{h}_{ij}, \mathrm{PE}(n)\right) + \frac{\sqrt{1-\bar{\alpha}_{n-1}}}{\sqrt{1-\bar{\alpha}_n}}\hat{\mathbf{x}}_{ij}^{t,(n)}, \tag{4}$$

where $\bar{\alpha}_n = \prod_{n'=1}^{n}\alpha_{n'}$ and $\alpha_n = 1-\beta_n$. At each reverse step n, $\mathrm{MLP}_{\mathrm{dif}}$ takes the noised embedding $\hat{\mathbf{x}}_{ij}^{t,(n)}$ and the review guidance $\mathbf{h}_{ij}$ as input and predicts the denoised embedding $\hat{\mathbf{x}}_{ij}^{t,(0)}$. $\bar{\alpha}_n$, α_n, and $\beta_n \in \mathbb{R}$ denote parameters that control the noise scale in the forward process at step n, determining the weights of $\hat{\mathbf{x}}_{ij}^{t,(n)}$ and the predicted $\hat{\mathbf{x}}_{ij}^{t,(0)}$ in the reverse process. We set β_n following a linear noise scheduling [4, 12, 28], linearly increasing from $\beta_1 = 0.0001$ to $\beta_N = 0.02$. By repeating the above update for N steps, we finally obtain $\hat{\mathbf{x}}_{ij}^{t,(0)}$ (*i.e.*, $\hat{\mathbf{x}}_{ij}^t$) from the initial noisy embedding $\hat{\mathbf{x}}_{ij}^{t,(N)}$.

3.2.3 Rating predictor. To predict the rating $\hat{r}_{ij}^t$ that the source-only user $u_i^{s\setminus o}$ would give to the target-domain item v_j^t, we input $\hat{\mathbf{x}}_{ij}^t$ to an MLP:

$$\hat{r}_{ij}^t = \mathrm{MLP}_{\mathrm{rec}}\left(\hat{\mathbf{x}}_{ij}^t\right), \tag{5}$$

where $\mathrm{MLP}_{\mathrm{rec}}(\cdot)$ has its own learnable parameters, independent of those of $\mathrm{MLP}_{\mathrm{dif}}(\cdot)$ in (C2).

3.3 Training Procedure

In this section, we describe in detail the two training stages of DIREC: (T1) pre-training and (T2) fine-tuning.

3.3.1 Pre-training. The goal of (T1) is to enable DIREC to better understand the distribution of review embeddings in the target domain before fine-tuning. To this end, we train the diffusion model on target-domain review embeddings $\mathcal{X}^{t\setminus o} = \{\mathbf{x}_{ij}^t \mid u_i^{t\setminus o} \in \mathcal{U}^t \setminus \mathcal{U}^o\}$ from target-only users (*i.e.*, $u_i^{t\setminus o} \in \mathcal{U}^t \setminus \mathcal{U}^o$).

For training, the diffusion model first applies the forward process for n steps by adding a Gaussian noise to the review embedding $\mathbf{x}_{ij}^t$, producing a noised review embedding $\mathbf{x}_{ij}^{t,(n)}$. Here, n is randomly sampled from $\{1,\dots,N\}$ at each training step. Thanks to the reparameterization trick [15], this n-step noising can be expressed in a

single step as follows:

$$q\left(\mathbf{x}_{ij}^{t,(n)} \mid \mathbf{x}_{ij}^{t,(0)}\right) = \mathcal{N}\left(\mathbf{x}_{ij}^{t,(n)};\ \sqrt{\bar{\alpha}_n}\,\mathbf{x}_{ij}^{t,(0)},\ (1-\bar{\alpha}_n)\mathbf{I}_d\right). \tag{6}$$

Accordingly, we can sample the noised review embedding as follows:

$$\mathbf{x}_{ij}^{t,(n)} = \sqrt{\bar{\alpha}_n}\,\mathbf{x}_{ij}^{t,(0)} + \sqrt{1-\bar{\alpha}_n}\,\boldsymbol{\epsilon}, \tag{7}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.

Given $\mathbf{x}_{ij}^{t,(n)}$, we train $\mathrm{MLP}_{\mathrm{dif}}(\cdot)$ to predict the noise-free embedding $\hat{\mathbf{x}}_{ij}^{t,(0)}$ (*i.e.*, the review embedding $\hat{\mathbf{x}}_{ij}^{t}$) by minimizing the following loss function $\mathcal{L}_{\text{p-dif}}$:

$$\mathcal{L}_{\text{p-dif}} = \frac{1}{|\mathcal{X}^{t\setminus o}|} \sum_{\mathbf{x}_{ij}^{t} \in \mathcal{X}^{t\setminus o}} \left\| \mathbf{x}_{ij}^{t} - \mathrm{MLP}_{\mathrm{dif}}\left(\mathbf{x}_{ij}^{t,(n)}, \mathbf{0}, \mathrm{PE}(n)\right)\right\|_2^2, \tag{8}$$

where $\mathbf{0} \in \mathbb{R}^d$ denotes the zero vector. Since $\mathrm{MLP}_{\mathrm{dif}}(\cdot)$ generates $\hat{\mathbf{x}}_{ij}^{t}$ without any conditioning signal, it learns the overall distribution of target-domain review embeddings.

We also train the rating predictor $\mathrm{MLP}_{\mathrm{rec}}(\cdot)$ to learn the mapping from generated target-domain review embeddings $\hat{\mathbf{x}}_{ij}^{t}$ to ratings r_{ij}^{t}. To this end, we optimize the Mean Squared Error (MSE) [35] loss $\mathcal{L}_{\text{p-rec}}$ by predicting the rating $\hat{r}_{ij}^{t}$ from the generated review embedding $\hat{\mathbf{x}}_{ij}^{t}$:

$$\mathcal{L}_{\text{p-rec}} = \frac{1}{|\mathcal{R}^{t\setminus o}|} \sum_{r_{ij}^{t} \in \mathcal{R}^{t\setminus o}} \left(r_{ij}^{t} - \mathrm{MLP}_{\mathrm{rec}}\left(\hat{\mathbf{x}}_{ij}^{t}\right)\right)^2, \tag{9}$$

where $\mathcal{R}^{t\setminus o}$ denotes the set of observed ratings on target-domain items v_j^t provided by target-only users $u_i \in \mathcal{U}^{t\setminus o}$. The final pre-training loss of DIREC is defined as follows:

$$\mathcal{L}_{\text{p}} = \lambda\,\mathcal{L}_{\text{p-dif}} + (1-\lambda)\,\mathcal{L}_{\text{p-rec}}, \tag{10}$$

where the hyperparameter λ controls the trade-off between $\mathcal{L}_{\text{p-dif}}$ and $\mathcal{L}_{\text{p-rec}}$. We provide a sensitivity analysis for λ in Section 4.

3.3.2 Fine-tuning. The goal of (T2) is to fine-tune the diffusion model pre-trained in (T1) so that it (i) generates a target-domain review embedding $\hat{\mathbf{x}}_{ij}^{t}$ for each overlapping user–target-domain item pair (u_i^o, v_j^t), optionally conditioned on the review guidance $\mathbf{h}_{ij}$, and (ii) predicts the corresponding rating $\hat{r}_{ij}^{t}$ based on $\hat{\mathbf{x}}_{ij}^{t}$. During fine-tuning, we randomly drop the review guidance $\mathbf{h}_{ij}$ with probability p to encourage the model to learn only the overall target-domain review-embedding distribution, following [13, 37, 40].

In (T2), we first obtain the review guidance $\mathbf{h}_{ij}$ via (C1) a review guidance encoder. We then apply the forward process, as in (T1), to obtain the noised review embedding $\mathbf{x}_{ij}^{t,(n)}$. Conditioned on the review guidance $\mathbf{h}_{ij}$ with probability $1-p$, we fine-tune the diffusion model to generate the target-domain review embedding $\hat{\mathbf{x}}_{ij}^{t}$ by optimizing the following loss function $\mathcal{L}_{\text{f-dif}}$:

$$\mathcal{L}_{\text{f-dif}} = \frac{1}{|\mathcal{X}^{o}|} \sum_{\mathbf{x}_{ij}^{t} \in \mathcal{X}^{o}} \left\| \mathbf{x}_{ij}^{t} - \mathrm{MLP}_{\mathrm{dif}}\left(\mathbf{x}_{ij}^{t,(n)}, \tilde{\mathbf{h}}_{ij}, \mathrm{PE}(n)\right)\right\|_2^2, \tag{11}$$

$$\tilde{\mathbf{h}}_{ij} = \begin{cases} \mathbf{0} & \text{with probability } p, \\ \mathbf{h}_{ij} & \text{with probability } 1-p. \end{cases} \tag{12}$$

where $\mathbf{x}_{ij}^{t} \in \mathcal{X}^{o}$ denotes the observed review embedding written by an overlapping user u_i^o for a target-domain item v_j^t. Since $\mathrm{MLP}_{\mathrm{dif}}(\cdot)$ is pre-trained on the target-domain review-embedding distribution, it could serve as an effective backbone that can be fine-tuned for the accurate generation of $\hat{\mathbf{x}}_{ij}^{t}$ conditioned on $\mathbf{h}_{ij}$.

Next, we fine-tune the rating predictor $\mathrm{MLP}_{\mathrm{rec}}(\cdot)$ to minimize the MSE loss $\mathcal{L}_{\text{f-rec}}$ between the predicted rating $\hat{r}_{ij}^{t}$ obtained from the generated review embedding $\hat{\mathbf{x}}_{ij}^{t}$ and the true rating r_{ij}^{t}:

$$\mathcal{L}_{\text{f-rec}} = \frac{1}{|\mathcal{R}^{o}|} \sum_{r_{ij}^{t} \in \mathcal{R}^{o}} \left(r_{ij}^{t} - \mathrm{MLP}_{\mathrm{rec}}\left(\hat{\mathbf{x}}_{ij}^{t}\right)\right)^2, \tag{13}$$

where $\mathcal{R}^{o}$ denotes the set of observed target-domain ratings assigned by overlapping users u_i^o to target-domain items v_j^t. The final fine-tuning loss $\mathcal{L}_{\text{f}}$ of DIREC is defined as follows:

$$\mathcal{L}_{\text{f}} = \lambda\,\mathcal{L}_{\text{f-dif}} + (1-\lambda)\,\mathcal{L}_{\text{f-rec}}. \tag{14}$$

Here, we use the same λ as in pre-training (Eq. 10).

Table 2: Dataset statistics for cross-domain recommendation. Over.U. and Over.Inter. denote the numbers of overlapping users and their interactions in each domain, respectively.

Scenarios	Users	Over.U.	Inter.	Over.Inter.	Items
Movie (source)	123,960	18,031	1,697,533	562,311	50,052
Music (target)	75,258		1,097,592	468,098	64,443
Book (source)	603,668	37,388	8,898,041	1,254,288	367,982
Movie (target)	123,960		1,697,533	792,319	50,052
Book (source)	603,668	16,738	8,898,041	638,404	367,982
Music (target)	75,258		1,097,592	380,675	64,443

4 Evaluation

In this section, we conduct experiments to evaluate the effectiveness of DIREC. We design our experiments to answer the following key evaluation questions (EQs):

- (EQ1): Does DIREC predict ratings more accurately than state-of-the-art CDRSs?
- (EQ2): Does generating target-domain review embeddings help improve rating prediction accuracy?
- (EQ3): Is target-aware source attention effective for improving rating prediction accuracy?
- (EQ4): Does pre-training the diffusion model on reviews from target-only users help improve rating prediction accuracy?
- (EQ5): How sensitive is the accuracy of DIREC to the hyperparameters H, N, and λ?

4.1 Evaluation Setup

Datasets. For evaluation, we prepare datasets by following the protocol used in previous CDRSs [18, 34, 37, 45]. We use three real-world datasets covering different domains: the Amazon review datasets[2], *i.e.*, Book, CDs and Vinyl (Music), and Movie. Each dataset follows a 5-core setting ensuring that every user and every item have at least five interactions, and provides both a rating and a review for every observed user–item interaction. Following prior work exactly [18, 34, 37, 45], we construct three cross-domain scenarios: Movie→ Music, Book→Movie, and Book→Music. For each scenario, we randomly select 20%, 50%, and 80% of overlapping users and mask all of their target-domain interactions (reviews and ratings) to simulate new users in the target domain; we treat these users as test users. We split the remaining overlapping users into a

[2]https://jmcauley.ucsd.edu/data/amazon/

Table 3: Accuracies of nine competitors and DIREC on three CDR scenarios. The improvements of DIREC over the best competitors are all statistically significant (p-value ≤ 0.001).

Methods	Movie → Music						Book → Movie						Book → Music					
	20%		50%		80%		20%		50%		80%		20%		50%		80%	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
LightGCN	3.8942	3.7453	3.8821	3.7299	3.8841	3.7345	3.7469	3.5538	3.7692	3.5821	3.7430	3.5519	3.9139	3.7724	3.9126	3.7735	3.9155	3.7765
DRRNN	1.2452	1.0342	1.4704	1.1961	1.6835	1.4890	1.1630	0.9579	1.1853	0.9871	1.3635	1.1308	1.2310	1.0678	1.4742	1.2040	1.7838	1.5784
EMCDR	1.9667	1.6544	1.9958	1.6146	2.5639	2.1525	1.4224	1.1378	1.5271	1.2090	1.6831	1.3282	2.0781	1.7483	2.1532	1.8384	2.6439	2.2989
RC-DFM	1.5465	1.3936	1.5629	1.4059	1.5993	1.4123	1.5489	1.3764	1.8955	1.7170	1.8459	1.6647	1.4185	1.2654	1.6975	1.5487	1.6390	1.4855
PTUPCDR	1.6681	1.2528	1.7668	1.2988	2.2843	1.6445	1.3668	1.0727	1.4569	1.1189	1.5990	1.2052	1.7333	1.3131	1.7632	1.3188	2.2257	1.6385
DiffCDR	1.4956	1.1052	1.7032	1.2234	2.2168	1.5685	1.2328	0.9516	1.3499	1.0328	1.4416	1.0552	1.5751	1.1571	1.6446	1.1930	2.1654	1.5723
JPEDET	1.3292	0.9538	1.5824	1.1432	2.1622	1.6633	1.2147	0.8926	1.3272	0.9896	1.6337	1.2198	1.4623	1.0452	1.6951	1.2334	2.1381	1.6160
DMCDR	1.0608	0.7609	1.0668	0.8116	1.0575	0.8166	1.2308	0.8732	1.1435	0.8898	1.1528	0.8893	1.0278	0.7878	1.0408	0.8041	1.0376	0.7877
PFCDR	1.0555	0.7545	1.0597	0.7991	1.1096	0.8092	1.0781	0.8158	1.0904	0.8112	1.1006	0.8427	1.0390	0.7717	1.0382	0.7775	1.0683	0.7926
DIREC	**0.9858**	**0.7013**	**0.9939**	**0.6934**	**0.9942**	**0.7036**	**1.0595**	**0.7748**	**1.0572**	**0.7641**	**1.0664**	**0.7825**	**0.9943**	**0.6919**	**0.9873**	**0.6829**	**0.9872**	**0.6759**
Gain (%)	6.60	7.05	6.21	13.23	5.99	13.05	1.73	5.03	3.04	5.81	3.11	7.14	3.26	10.34	4.90	12.17	4.86	14.19

Table 4: Top-K performance of competitors and DIREC on three CDR scenarios.

Scenarios	Metric	LightGCN	JPEDET	PFCDR	DIREC
Movie → Music	**HR@10**	0.1060	0.0871	0.1010	**0.1562**
	NDCG@10	0.0486	0.0369	0.0448	**0.0741**
Book → Movie	**HR@10**	0.1063	0.0956	0.1097	**0.1968**
	NDCG@10	0.0487	0.0384	0.0497	**0.0908**
Book → Music	**HR@10**	0.1021	0.0809	0.0956	**0.1201**
	NDCG@10	0.0462	0.0348	0.0432	**0.0542**

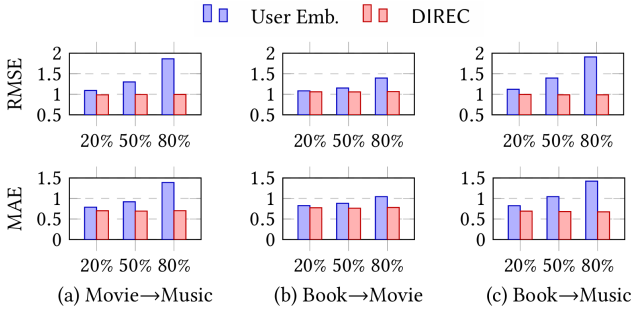


Figure 3: Effectiveness of target-domain review embedding generation. The x-axis denotes the ratio of test users, and the y-axis denotes the metric value.

training set (90%) and a validation set (10%). Table 2 summarizes the statistics of datasets.

Competitors. To verify the effectiveness of DIREC, we compare DIREC with nine competitors. We include two single-domain recommenders (*i.e.*, LightGCN [11] and DRRNN [36]). For these two single-domain recommenders, we use the same train/validation/test splits as CDRSs, but merge the interactions from both domains into a single interaction set. We also include seven state-of-the-art CDRSs (*i.e.*, EMCDR [26], RC-DFM [8], PTUPCDR [45], DiffCDR [37], JPEDET [19], DMCDR [18] and PFCDR [34]).

Following [8, 18, 19, 26, 34, 36, 45], we focus on rating prediction tasks; we evaluate the performance of DIREC and competitors using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) as accuracy metrics. All reported results are the averages of five independent runs conducted with different random seeds.

Implementation Details. All experiments are conducted on an Ubuntu 22.04 server equipped with NVIDIA RTX A6000 GPUs (48GB VRAM) and 256GB RAM, implemented using PyTorch 2.7.0. We set the number of layers of MLP_{dif} and MLP_{rec} to 6 and 4, respectively. We apply a dropout with rate 0.1 to the MLP layers and use 4 heads for multi-head attention. The sentence-transformer used for review-encoding has an embedding dimension of $d = 768$, and its parameters are frozen. Following [13, 37], we set the condition drop probability p to 0.1. We tune $H \in \{10, 20, 30\}$, $N \in \{500, 1000, 1500\}$, and $\lambda \in \{0.1, 0.2, \ldots, 0.9\}$. We use a batch size of 512 and the OneCycle learning rate schedule [29], and train the model using AdamW [22] with a weight decay of 0.0001. We employ early stopping based on the validation loss with a patience of 10 epochs. For competitors, we use the best hyperparameter settings reported in their respective papers. Following DDIM [30], we accelerate the inference of the diffusion model by sampling 100 uniformly spaced diffusion steps instead of all N steps, making the inference time independent of N. Due to space limitations, we omit the inference time of DIREC here, providing it in Appendix A.

4.2 Results

(EQ1) Comparison with Competitors. We compare the rating prediction accuracy of DIREC with those of nine competitors to demonstrate the effectiveness of DIREC. Table 3 presents the results where we highlight the best accuracies with **boldface** and the second-best accuracies with underlines. Also, 'Gain' indicates the degree of accuracy improvements by DIREC over the *best* competitors. Our findings are summarized as follows:

- Most importantly, DIREC *consistently* and *significantly* outperforms *all* competitors in *all* scenarios for *all* metrics. Moreover, the gain becomes larger as the ratio of test users among overlapping users increases. Specifically, when 80% of the overlapping users are used as test users, DIREC outperforms the best competitors (*i.e.*, PFCDR and DMCDR) by 13.05%, 7.14%, and 14.19% in MAE on Movie→Music, Book→Movie, and Book→Music, respectively.
- Comparing the two single-domain recommenders (LightGCN and DRRNN), we find that DRRNN, which initializes user and item embeddings using reviews, substantially outperforms LightGCN, which randomly initializes embeddings and relies solely on CF signals, across all scenarios. This suggests that CF signals are

Table 5: Effects of real-generated review embedding similarity on cross-domain recommendation performance

Scenarios	Metric	No Attn.	Self Attn.	No Pre.	DIREC
Movie → Music	**RMSE**	1.0333	1.0047	1.0103	**0.9858**
	L2 dist.	2.6797	1.7218	2.9563	**1.6605**
Book → Movie	**RMSE**	1.1033	1.0666	1.0855	**1.0595**
	L2 dist.	3.4200	**1.4373**	3.5076	1.4480
Book → Music	**RMSE**	1.0175	1.0072	0.9971	**0.9943**
	L2 dist.	1.3087	1.3348	1.5823	**1.2760**

domain-specific and difficult to generalize across domains without explicit transfer learning, while reviews provide transferable semantic information that facilitates cross-domain transfer.

- The review-based CDRSs (JPEDET and RC-DFM) do not consistently outperform all non-review-based CDRSs, despite leveraging reviews. We note that these methods, similar to non-review-based CDRSs, still predict ratings without explicitly learning to generate target-domain review embeddings. As a result, they may fail to capture fine-grained preference signals for a given user–item pair revealed in reviews, leading to unsatisfactory recommendation accuracy. In contrast, DIREC generates a review embedding for each user–item pair and predicts the rating based on the generated embedding, enabling it to capture fine-grained user–item preferences more accurately.

While many CDR works mainly report rating prediction metrics [8, 18, 19, 26, 34, 36, 45], we also evaluate top-k ranking performance to assess our method from multiple perspectives. Following a standard sampled-ranking protocol, we rank 1 ground-truth positive item against 100 sampled negative items per test case. We compare DIREC with PFCDR (the strongest competitor in rating prediction), JPEDET (a representative review-based CDRS), and LightGCN (a widely used top-k recommender). As shown in Table 4, DIREC consistently achieves the best ranking performance by a large margin; among the baselines, LightGCN performs best, followed by PFCDR and then JPEDET. These results show that the benefit of DIREC extends beyond rating prediction and also translates into stronger top-k recommendation quality, further supporting the effectiveness of generating target-domain review embeddings. These results indicate that DIREC also performs strongly on top-k ranking metrics, further supporting the effectiveness of generating target-domain review embeddings.

(EQ2) Effectiveness of the Review-Embedding Generation. We design DIREC to first generate a target-domain review embedding and then predict the rating based on the generated embedding. Consequently, DIREC learns fine-grained user–item preference signals that may not be fully reflected in a scalar rating. To verify whether review-embedding generation is beneficial in improving rating prediction accuracy, we compare the rating prediction accuracy of DIREC with that of a variant that generates a *user embedding* in the target domain, as in conventional CDRSs, instead of generating a review embedding. Specifically, this variant constructs a condition embedding that represents the user's overall preference in the source domain by applying self-attention to the user's source-domain review embeddings, followed by mean pooling. Conditioned on this embedding, it trains a diffusion model to generate the user embedding in the target-domain, which we define as the

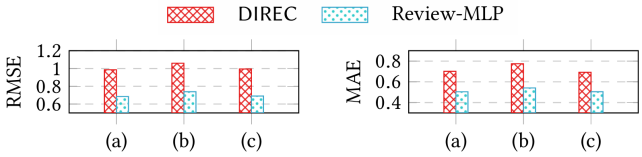


Figure 4: Performance Comparison Between DIREC and Review-MLP on (a) Movie → Music, (b) Book → Movie, and (c) Book → Music.

average embedding of the user's target-domain reviews. Finally, following [26, 34, 37, 45], it predicts the rating using the dot product between the generated user embedding in the target-domain and the item embedding in the target-domain. Similar to DIREC, it is trained in two stages: pre-training on target-only users, followed by fine-tuning on overlapping users.

Figure 3 illustrates the results. First, we observe that DIREC *consistently* and *substantially* outperforms the variant (User Emb.) across *all* scenarios and *all* metrics. Next, we find that the variant performs slightly better than existing review-based CDRSs (JPEDET and RC-DFM). These results suggest that diffusion is more effective in transferring information from the source domain to the target domain than prior approaches. Meanwhile, the results also highlight the limitation of predicting ratings solely via embedding matching between a user and an item *without explicitly learning to generate review embeddings*.

To further verify whether review-embedding generation is beneficial for rating prediction, we conduct two additional experiments.
(1) How generated review-embedding quality relates to rating accuracy. DIREC is designed to generate accurate target-domain review embeddings via two key designs: target-aware source attention and pretraining on target-only users. We evaluate two guidance variants (replacing target-aware source attention) and one training variant (removing target-only pretraining): (i) Avg, which replaces target-aware source attention by aggregating the user's SREs via averaging; (ii) Self Attn., which applies self-attention [33] over the user's SREs (queries/keys/values are all set to SREs) followed by mean pooling; and (iii) No Pre., which removes pretraining on target-only users. For each method, we compare rating prediction accuracy (RMSE) and the distance between the generated review embedding and the real target-domain review embedding for the same user–item pair (ℓ_2 distance). As shown in Table 5, DIREC produces review embeddings that are closest to real target review embeddings while also achieving the best rating prediction accuracy, suggesting that improving review-embedding generation quality helps improve rating prediction.
(2) Predicting ratings from real target review embeddings. To assess how informative review embeddings are for modeling user–item preferences, we additionally train a simple MLP regressor, Review-MLP, that predicts the rating solely from the real target-domain review embedding. As shown in Figure 4, Review-MLP achieves substantially better accuracy than DIREC despite its simplicity, indicating that real review embeddings contain highly informative preference signals for a given user–item pair. This result highlights the importance of generating target-domain review embeddings and predicting ratings from those embeddings.

(EQ3) Effectiveness of Target-aware Source Attention. To construct review guidance, we employ *target-aware source attention*,

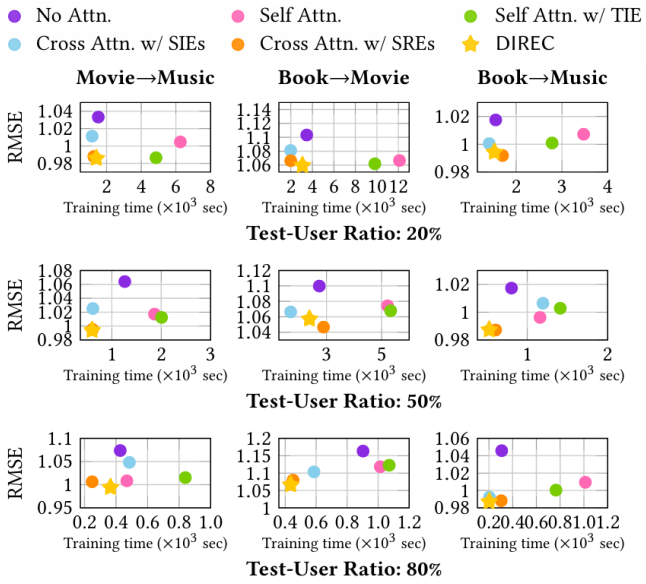


Figure 5: Effectiveness and efficiency of the target-aware source attention.

where the target-domain item embedding (TIE) serves as a *query*, the source-domain item embeddings (SIEs) of items interacted with the user serve as *keys*, and the source-domain review embeddings (SREs) written by the user for those items serve as *values*. To validate the effectiveness of our target-aware source attention, we compare DIREC with the following five variants that differ only in how they construct review guidance:

- **Avg.**: a variant that aggregates the user's SREs by *averaging*.
- **Cross Attn. w/ SIEs**: a variant that performs *cross-attention* [25] with the query set to TIE (same as DIREC) and both keys and values set to the SIEs of items interacted with the user; note that SIEs are used only as attention keys in DIREC.
- **Cross Attn. w/ SREs**: a variant that performs CA with the query set to TIE (same as DIREC) and both keys and values set to the SREs of the user; note that SREs are used only as attention values in DIREC.
- **Self Attn.**: a variant that applies *self-attention* to the SREs of the user, where the queries, keys, and values are all set to the SREs, followed by mean pooling.
- **Self Attn. w/ TIE**: a variant that applies SA to the SREs of the user and the TIE, where the queries, keys, and values are all set to these embeddings, followed by mean pooling.

Figure 5 reports the RMSE (y-axis) and the training (fine-tuning) time (x-axis) of the variants and DIREC. Due to space limitations, we provide MAE results in Appendix A. In terms of accuracy, DIREC achieves the best performance. Next, the Self-Attn.-based variants (Self-Attn. and Self-Attn. w/ TIE) and Cross-Attn. w/ SREs follow DIREC with comparable accuracy, while Cross-Attn. w/ SIEs still outperforms Avg. In terms of training time, Avg. and the Cross-Attn.-based variants, including DIREC, are comparable, while the Self-Attn.-based variants require the longest training time. These results lead to two important observations. First, compared to Cross-Attn. w/ SIEs, DIREC and Cross-Attn. w/ SREs consistently achieve higher accuracy, suggesting that SREs are a better choice as *values*

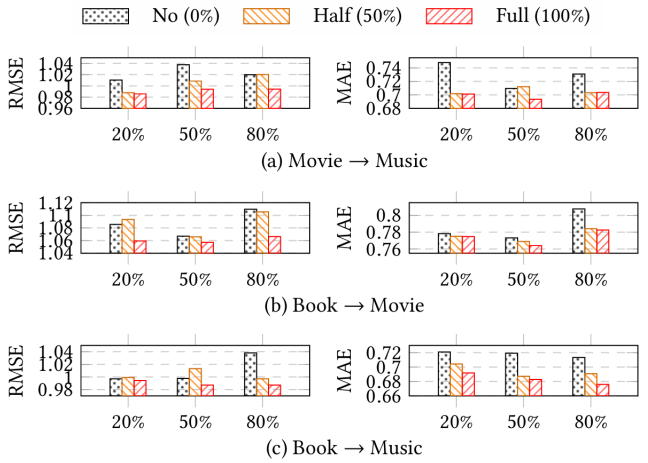


Figure 6: Prediction accuracy under different pre-training proportions of target-only user reviews.

for constructing user-specific review guidance because they capture the user's preferences more precisely than SIEs. Second, compared to Cross-Attn. w/ SREs, DIREC further improves accuracy by using SIEs as *keys* to identify source-domain items most similar to the given target-domain item, achieving the best accuracy while requiring substantially less training time than the Self-Attn.-based variants. In summary, by carefully designing the query, key, and value in the attention mechanism, DIREC generates the review guidance both effectively and efficiently.

(EQ4) Effectiveness of the Target-only User Review Pre-training. We pre-train the diffusion model with the review embeddings from target-only users to accurately model the distribution of target-domain review embeddings. To validate the effectiveness of this pre-training, we vary the proportion of target-only users' review embeddings used for pre-training in DIREC (*i.e.*, 0%, 50%, and 100%) and compare the prediction accuracy.

Figure 6 shows the results. Overall, using *all* review embeddings from target-only users for pre-training (Full) achieves the highest prediction accuracy, followed by 50% (Half), while 0% (No) performs the worst. Notably, the 100% pre-training setting exhibits the smallest variance in accuracy across different test-user ratios (20%, 50%, and 80%). These results indicate that pre-training on target-only users' review embeddings helps the diffusion model better capture the target-domain review-embedding distribution. Consequently, DIREC maintains robust prediction accuracy even when fine-tuned with limited overlapping user data.

(EQ5) Parameter Sensitivity Analysis. DIREC requires setting three hyperparameters: (i) the number of sampled source-domain items H to construct the review guidance, (ii) the number of diffusion steps N, and (iii) the weighting factor λ that balances the diffusion loss $\mathcal{L}_{\text{p-dif}}$ (resp. $\mathcal{L}_{\text{f-dif}}$) and the rating-prediction loss $\mathcal{L}_{\text{p-rec}}$ (resp. $\mathcal{L}_{\text{f-rec}}$). To analyze the impact of these hyperparameters in DIREC, we perform a sensitivity analysis for them. Specifically, we vary $H \in \{10, 20, 30\}$, $N \in \{500, 1000, 1500\}$, and $\lambda \in \{0.1, 0.2, \ldots, 0.9\}$, and compare their prediction accuracies. Due to space limitations, we report the RMSE results in this section and provide the MAE results in Appendix A.

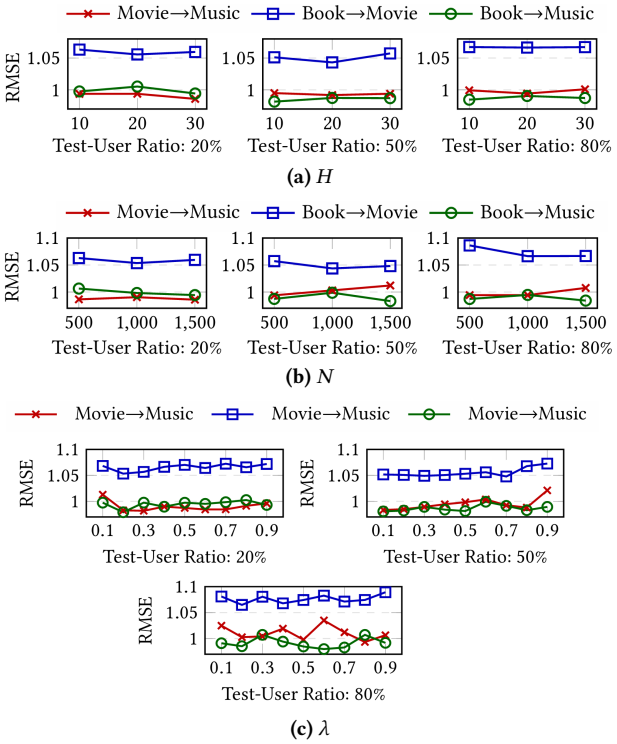


Figure 7: Hyperparameter sensitivity analysis of H, N, and λ.

Figure 7 illustrates the results. Overall, the prediction accuracy changes *only marginally* across the tested ranges of H, N, and λ, indicating that DIREC is insensitive to hyperparameter choices. For H, the accuracies are comparable across values, suggesting that only a small number of sampled source items (*e.g.*, $H = 10$) are sufficient to provide effective review guidance. Although $N = 1500$ slightly improves accuracy in some cases, the gain is marginal; and we observe comparable accuracy with much fewer diffusion steps. Finally, the accuracy remains stable over a wide range of λ, suggesting that training is not sensitive to the relative weighting between the diffusion and rating-prediction losses.

5 Related Work

In this section, we briefly review previous cross-domain recommendation systems (CDRSs). CDRSs can be classified into two categories depending on their objectives: (1) providing recommendations in the target domain for *target-domain new users* who have interactions only in the source domain [8, 18, 26, 34, 45]; and (2) providing recommendations in the target domain by leveraging users' interaction histories in both the target and source domains to alleviate data sparsity in the target domain [5, 9, 21, 38]. In this work, we focus on (1), *i.e.*, recommending for target-domain new users.

Previous studies have leveraged overlapping users to capture the relationships between user preferences in the source and target domains. They typically construct domain-specific user embeddings and learn a mapping function that transfers user embeddings from the source domain to the target domain [8, 18, 19, 26, 34, 37, 45]. For instance, EMCDR [26] utilizes matrix factorization to learn the embeddings in each domain and employs an MLP-based mapping function to bridge the embedding spaces of the two domains. PTUPCDR [45] employs a user-specific meta network to parameterize personalized bridge functions for user embedding transfer. DiffCDR [37] replaces conventional mapping networks with a diffusion model that generates target-domain user embeddings conditioned on source-domain user embeddings. Recently, PFCDR [34] distills privacy-friendly user embeddings (*i.e.*, user prototypes) from interactions in the source-domain via a *conditional model inversion*, and then transfers these prototypes to the target domain for recommendation.

Meanwhile, several CDR studies have leveraged review information to construct user and item embeddings [8, 19, 41, 43]. RC-DFM [8] deeply fuses ratings and reviews through stacked denoising autoencoders to learn user embeddings and then trains an MLP to map these embeddings across source- and target-domains. JPEDET [19] constructs the user's general preference embedding by jointly leveraging her reviews and ratings, and employs a dynamic embedding transportation mechanism to enable bidirectional transformation between the two domains. MuSiC [41] extracts domain-unified multimodal item and user embeddings via MLLMs/LLMs and trains a diffusion model that learns the target-domain embedding distribution from side users and the cross-domain transformation from overlapping users to generate target-domain user embeddings for cold-start recommendation.

However, existing CDRSs still focus on transferring user embeddings and predicting ratings via *embedding matching*, which can limit their ability to model fine-grained, item-specific preferences revealed in reviews. DIREC learns to generate target-domain review embeddings for each user–item pair, enabling more fine-grained preference modeling in the target domain.

6 Conclusion

In this paper, we claimed that learning to generate review embeddings is crucial for cross-domain recommendation. To this end, we proposed DIREC, a diffusion-based framework that generates a target-domain review embedding for each user–item pair and predicts the corresponding rating from the generated embedding. We further improve DIREC with two ideas: **(Idea 1)** target-aware source attention to construct review guidance aligned with the target-domain item and **(Idea 2)** pre-training on target-domain review embeddings from target-only users. Through extensive experiments, we show that (i) DIREC consistently outperforms nine state-of-the-art competitors, (ii) learning to generate the review embedding improves prediction accuracy, and (iii) our two ideas are essential to achieving the superior performance of DIREC.

Acknowledgments

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A Further Results

In this appendix, we report the inference time of DIREC and supplementary evaluation results for EQ3 and EQ5.

Inference Time of DIREC. We accelerate the inference of the diffusion model by sampling 100 uniformly spaced diffusion steps instead of all N steps. To verify the scalability of DIREC, we measure its inference time on each dataset. Table 6 reports the inference time of DIREC along with the number of test interactions for each dataset. The results show that the inference time of DIREC scales linearly with the number of test interactions. On average, DIREC requires only $\approx 4 \times 10^{-4}$ seconds to predict a single rating. These results demonstrate that DIREC is sufficiently scalable.

Table 6: Inference time of DIREC for each scenario

Scenario		Time (sec)	Test Inter.
Movie → Music	**20%**	38.7	94,202
	50%	95.5	237,811
	80%	154	379,177
Book → Movie	**20%**	65.8	156,408
	50%	171.5	393,418
	80%	270.1	638,071
Book → Music	**20%**	27.7	72,029
	50%	69.9	186,211
	80%	121.6	304,267

Further Results for EQ3. For EQ3, we compare DIREC with its variants that use different modules to construct the review guidance. While Figure 5 reports the RMSE and the training time, Figure 8 shows the MAE (y-axis) and the training (fine-tuning) time of DIREC and its variants. The results are consistent with those in Figure 5, DIREC achieves the lowest MAE with the shortest training time.

Further Results for EQ5. For EQ5, we analyze how the accuracy of DIREC changes with different parameter settings for H, N, and λ. While Figure 7 reports the RMSE, Figure 9 shows the MAE for all settings. The results are consistent with those in Figure 7, the prediction error of DIREC is insensitive to hyperparameter choices.

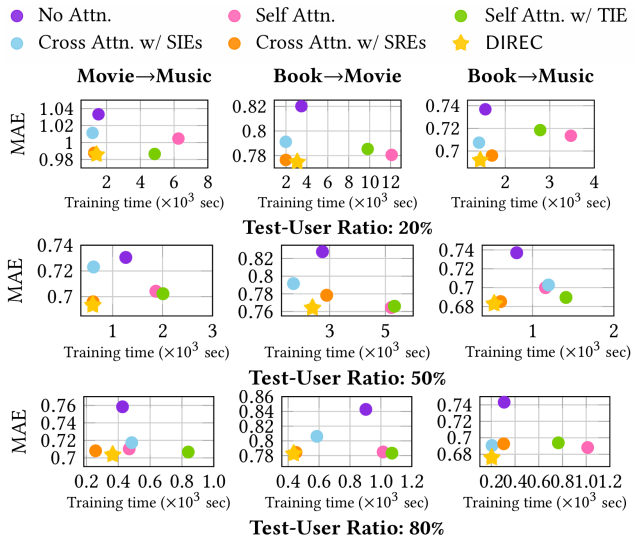


Figure 8: Effectiveness and efficiency of the target-aware source attention.

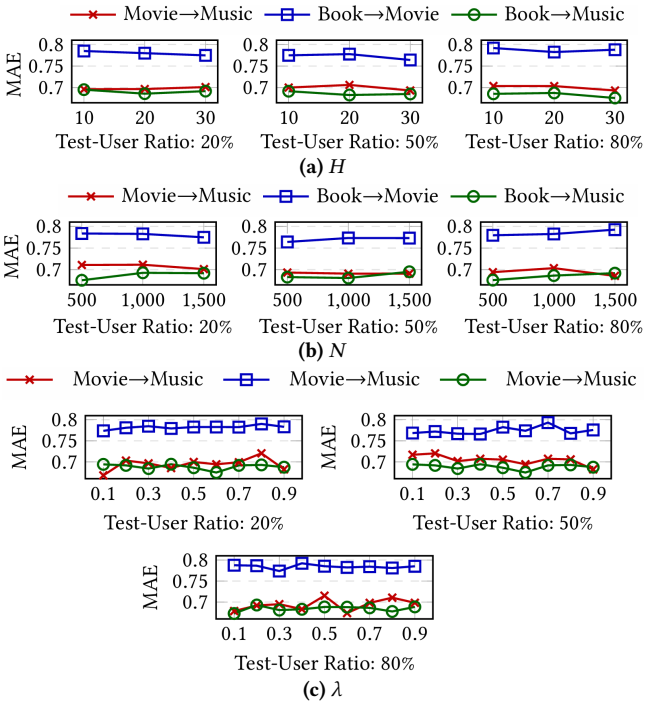


Figure 9: Hyperparameter sensitivity analysis of H, N, and λ.